

Uniform Distribution

23 March 2023 13:19

What is Uniform Distribution and it's types

In probability theory and statistics, a uniform distribution is a probability distribution where all outcomes are equally likely within a given range. This means that if you were to select a random value from this range, any value would be as likely as any other value.

Types



Denoted as

$$X \sim U(a, b) \rightarrow \text{parameters}$$

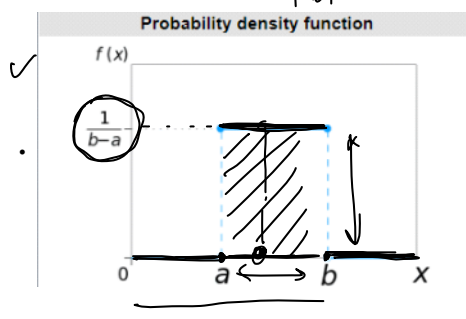
$a \leftrightarrow b$
 lower higher

Examples

- The height of a person randomly selected from a group of individuals whose heights range from 5'6" to 6'0" would follow a continuous uniform distribution.
- The time it takes for a machine to produce a product, where the production time ranges from 5 to 10 minutes, would follow a continuous uniform distribution.
- The distance that a randomly selected car travels on a tank of gas, where the distance ranges from 300 to 400 miles, would follow a continuous uniform distribution.
- The weight of a randomly selected apple from a basket of apples that weighs between 100 and 200 grams, would follow a continuous uniform distribution.

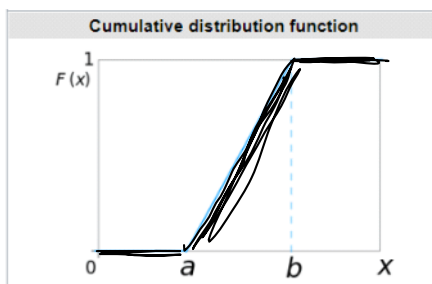
PDF CDF and Graphs

PDF



$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{for } x < a \text{ or } x > b \end{cases}$$

CDF

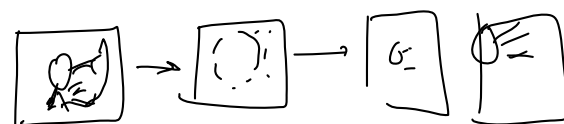


https://en.wikipedia.org/wiki/Continuous_uniform_distribution

- Skewness $\rightarrow 0 \rightarrow$ symmetric $\rightarrow$ Normal

Application in Machine learning and Data Science

- Random initialization** In many machine learning algorithms, such as neural networks and k-means clustering, the initial values of the parameters can have a significant impact on the final result. Uniform distribution is often used to randomly initialize the parameters, as it ensures that all values in the range have an equal probability of being selected.

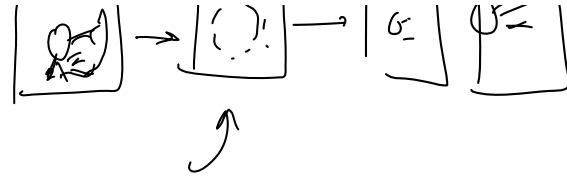


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b. **Sampling:** Uniform distribution can also be used for sampling. For example, if you have a dataset with an equal number of samples from each class, you can use uniform distribution to randomly select a subset of the data that is representative of all the classes.

c. **Data augmentation:** In some cases, you may want to artificially increase the size of your dataset by generating new examples that are similar to the original data. Uniform distribution can be used to generate new data points that are within a specified range of the original data.

d. **Hyperparameter tuning:** Uniform distribution can also be used in hyperparameter tuning, where you need to search for the best combination of hyperparameters for a machine learning model. By defining a uniform prior distribution for each hyperparameter, you can sample from the distribution to explore the hyperparameter space.



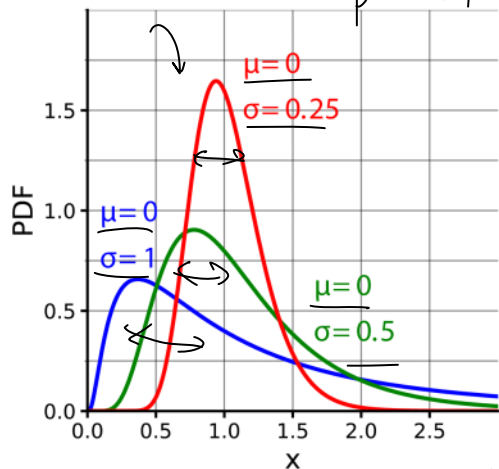
→ deep learning
CNN

Log Normal Distribution

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In probability theory and statistics, a lognormal distribution is a heavy tailed continuous probability distribution of a random variable whose logarithm is normally distributed.

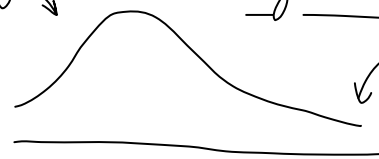
parameters $\rightarrow (\mu, \sigma)$



Examples

- The length of comments posted in Internet discussion forums follows a log-normal distribution.
- Users' dwell time on online articles (jokes, news etc.) follows a log-normal distribution.
- The length of chess games tends to follow a log-normal distribution.
- In economics, there is evidence that the income of 97%-99% of the population is distributed log-normally.

log normal $\leftarrow$ right skewed



$$\log(x) \sim N(\mu, \sigma)$$

$$X \sim \text{log Normal}$$

$$\log(x) \sim N(\mu, \sigma)$$

Age



$$\ln(27)$$

$$\ln(28)$$

$$\ln(25)$$

$$\ln(21)$$



insta, reddit, facebook, youtube

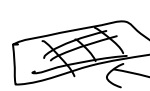
#words

video

$$C_1 \rightarrow 23$$

$$C_2 \rightarrow 41$$

$$C_3 \rightarrow 100$$



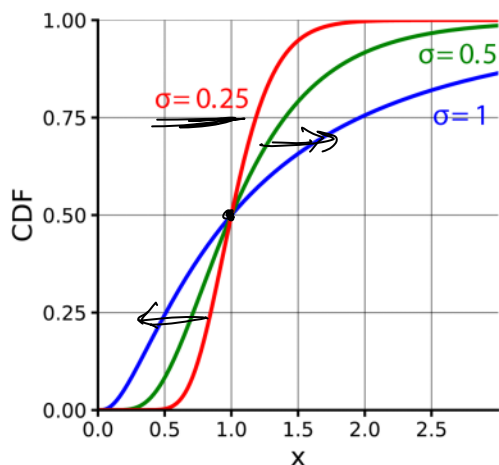
Denoted as

$$X \sim \text{lognormal}(\mu, \sigma) \rightarrow \ln(x) \sim N(\mu, \sigma)$$

PDF Equation

$$\frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}} \leftarrow \text{similar Normal dist.}$$

CDF

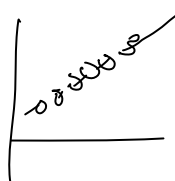
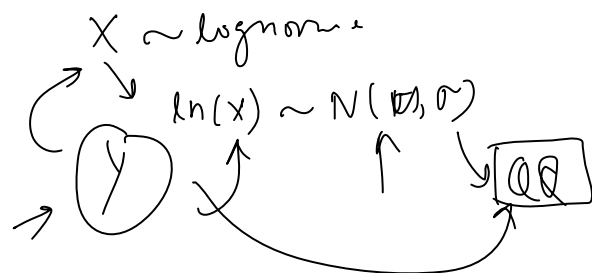


Skewness

skewed

How to check if a random variable is log normally distributed?

How to check if a random variable is log normally distributed?



Pareto Distribution

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Pareto Distribution

The Pareto distribution is a type of probability distribution that is commonly used to model the distribution of wealth, income, and other quantities that exhibit a similar power-law behaviour

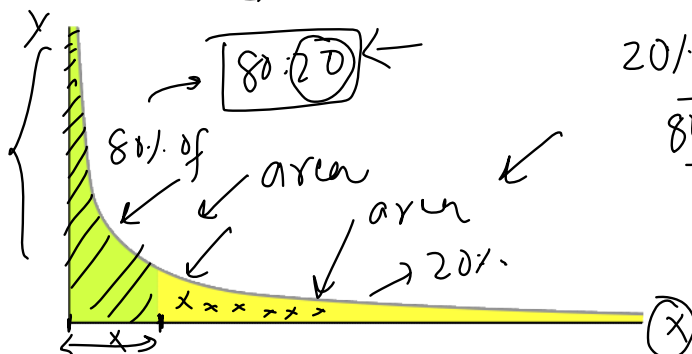
What is Power Law

In mathematics, a power law is a functional relationship between two variables, where one variable is proportional to a power of the other. Specifically, if y and x are two variables related by a power law, then the relationship can be written as:

$$y = k * x^a$$

$$y = k x^a$$

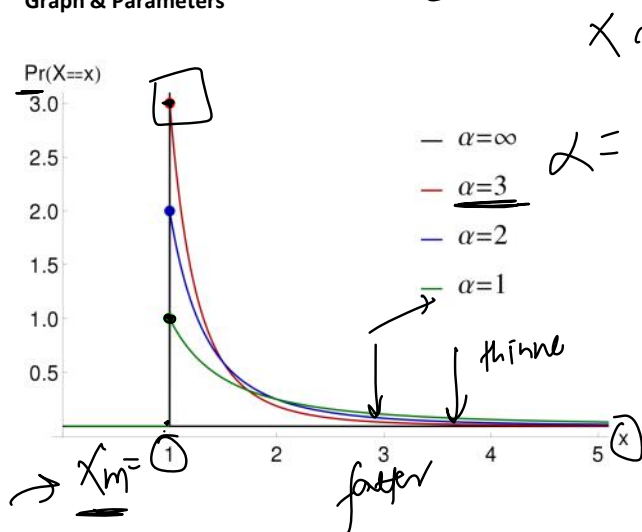
20% of x has 80% of entire



Vilfredo Pareto originally used this distribution to describe the allocation of wealth among individuals since it seemed to show rather well the way that a larger portion of the wealth of any society is owned by a smaller percentage of the people in that society. He also used it to describe distribution of income. This idea is sometimes expressed more simply as the Pareto principle or the "80-20 rule" which says that 20% of the population controls 80% of the wealth

↑ application

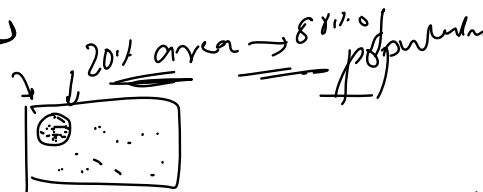
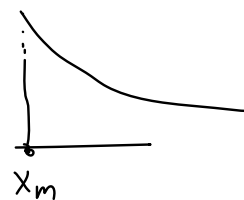
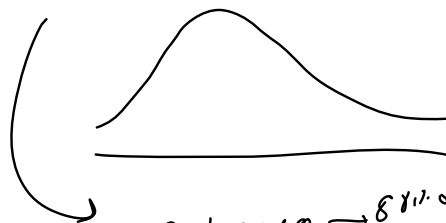
Graph & Parameters



$$X \sim \text{Pr}(\alpha)$$

$$\alpha = 1.16$$

$$\text{pdf} = \frac{\alpha x_m^\alpha}{x^{\alpha+1}} = p$$



Examples

- The sizes of human settlements (few cities, many hamlets/villages)
- File size distribution of Internet traffic which uses the TCP protocol (many smaller files, few larger ones)

80% of entire go

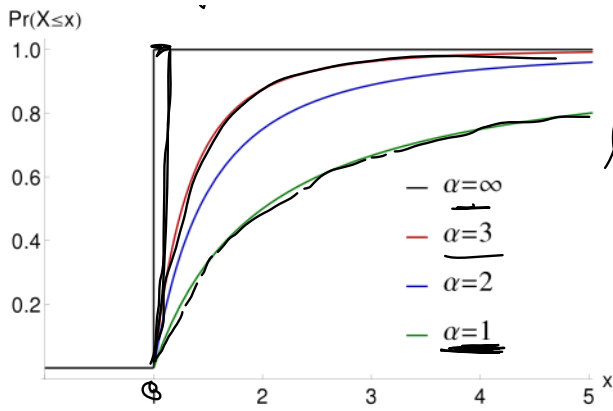
20% of 80%

20% of file

80%

CDF

$$\text{Pr}(X < x)$$

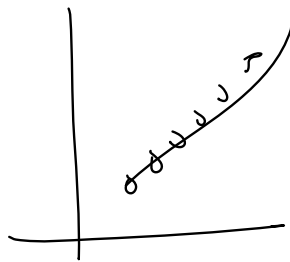


Skewness → skewed

[How to detect if a distribution is Pareto Distribution?]

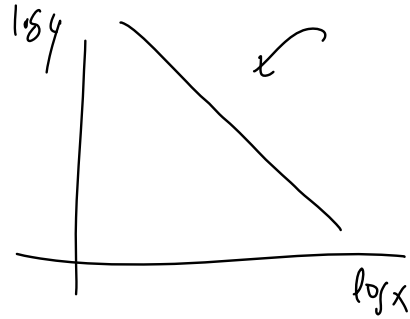
QQ plot

x, y →
↑
pareto



log (y) →
pareto
x log

$$y = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}$$



Transformations

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